
[162060] LOW-RESOURCE MACHINE TRANSLATION

SoSe 2026

LANGUAGE:	English	TEACHER:	Christian Schuler
PLACE:	C7.2 - 1.05	OFFICE:	D3 1 (Altbau) +1, Room 1.09
TIME:	16:15-17:45	E-MAIL:	christian.schuler@dfki.de
START:	14.04.2026	OBERBABO:	Simon Ostermann
CREDITS:	4 CP (R), 7 CP (R+H)		
WORKLOAD:	2 SWS		

COURSE DESCRIPTION

This seminar explores the challenges and possibilities of machine translation in contexts where data is scarce, incomplete, or entirely absent. Rather than focusing on standard, high-resource scenarios, the course examines what happens when core assumptions of modern NLP break down: when languages lack large corpora, established tools, or even a written tradition. Students will engage with both the technical and conceptual foundations of low-resource machine translation. Topics include data acquisition and corpus creation, evaluation under uncertainty, multilingual and transfer learning approaches, and strategies such as data augmentation, zero- and few-shot learning, and active learning. Alongside these methods, the course highlights the role of linguistic knowledge as a structured alternative to purely data-driven approaches. A central thread throughout the seminar is the critical examination of the broader context in which these systems are developed and deployed. Questions of ethics, bias, and responsibility are treated as integral to the field: whose languages are supported, how linguistic communities are represented, and what risks arise when systems are applied in sensitive or high-stakes domains. The course is intended for students with an interest in natural language processing, machine learning, and linguistic diversity, who want to understand both the limitations of current approaches and the strategies being developed to overcome them. It is particularly suited for those interested in research questions at the intersection of technology, language, and society.

SEMINAR, SEMINAR- WHAT EVEN IS A SEMINAR?

“A seminar is a form of (...) recurring meetings, focusing each time on some particular subject, in which everyone present is requested to participate. This is often accomplished through an ongoing Socratic dialogue (...) It is essentially a place where assigned readings are discussed, questions can be raised and debates can be conducted.” *

SEMINAR SCHEDULE

Each session has a pool of papers, and we plan to have two presentations per session. Students present a paper of their choice (under the constraint that we need to fill each session) from the session's paper pool and lead the class discussion.

Some words on grading: This seminar is meant to be as interactive as possible. Final grades will be based on students' presentations, term papers (optional), but also on participation and discussion in class.

The participants are expected to prepare for classes accordingly, by reading the relevant papers and also doing background reading, if necessary. Based on this preparation, the participants should be able to discuss the presented papers in depth and to understand relevant context during the discussion.

*<https://en.wikipedia.org/wiki/Seminar>

INTRODUCTION, COURSE OVERVIEW, AND ADMINISTRATIVE MATTERS

“What are we going to do here?”

Session	Date	Content	Paper	Student
Phase 1 - Warm-Up				
1	14.04.2026	Administrative Session & Paper Pool		
2	21.04.2026	Introductory Lecture		
3	28.04.2026	Lingering Questions (optional)		
4	05.05.2026	Lingering Questions (optional)		
5	12.05.2026	LREC (no meeting)		
Phase 2 - Student's Paper Presentations				
6	19.05.2026	Problem Space & Linguistic Foundations	(Kornai, 2013)	P.H. / Q?
7	26.05.2026	Ethics & Policy		A? / N?
8	02.06.2026	Data Engineering & Augmentation	(Riabi et al., 2023)	A.K.
9	09.06.2026	Sub-word Modeling & Representation		
10	16.06.2026	Evaluation & Quality Estimation		J? / N?
11	23.06.2026	Domain Adaptation & Moadlities		J?
12	30.06.2026	Learning Paradigms & Architectural Methods		F?
13	07.07.2026	Emerging Directions & Open Problems		F?
Phase 3 - Wrap-Up				
14	14.07.2026	Class Review & Synthesis		

PROBLEM SPACE & LINGUISTIC FOUNDATIONS

“What defines low-resource languages and why do they pose unique challenges for MT?”

Digital Language Death

Kornai (2013) | [URL](#)

Survey of Low-Resource Machine Translation

Haddow et al. (2022) | [URL](#)

Speech Translation and the End-to-End Promise: Taking Stock of Where We Are

Sperber and Paulik (2020) | [URL](#)

The State and Fate of Linguistic Diversity and Inclusion in the NLP World

Joshi et al. (2020) | [URL](#)

ELLORA: Enabling Low Resource Languages with Technology

Bali et al. (2019) | [URL](#)

Cyber-geography of languages: Part 1: method, results and focus on English

Pimienta and de Oliveira (2022) | [URL](#)

Decolonising Speech and Language Technology

Bird (2020) | [URL](#)

Additional Reading

Language Death

Crystal (2000) | [URL](#)

Atlas of the world's languages in danger

Moseley and Nicolas (2010) | [URL](#)

ETHICS & POLICY

“Who benefits, who is harmed, and what governance is required?”

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?

Bender et al. (2021) | [URL](#)

Challenges of Language Technologies for the Indigenous Languages of the Americas

Mager et al. (2018) | [URL](#)

Local Languages, Third Spaces, and other High-Resource Scenarios

Bird (2022) | [URL](#)

The State and Fate of Linguistic Diversity and Inclusion in the NLP World

Joshi et al. (2020) | [URL](#)

Decolonising Speech and Language Technology

Bird (2020) | [URL](#)

Not always about you: Prioritizing community needs when developing endangered language technology

Liu et al. (2022) | [URL](#)

DATA ENGINEERING & AUGMENTATION

“How to acquire, curate, or synthesize data under extreme scarcity?”

Europarl: A Parallel Corpus for Statistical Machine Translation
Koehn (2005) | [URL](#)

CCMatrix: Mining Billions of High-Quality Parallel Sentences on the Web
Schwenk et al. (2021) | [URL](#)

Findings of the WMT 2018 Shared Task on Parallel Corpus Filtering
Koehn et al. (2018) | [URL](#)

Seven Dimensions of Portability for Language Documentation and Description
Bird and Simons (2002) | [URL](#)

Massively Multilingual Sentence Embeddings for Zero-Shot Cross-Lingual Transfer and Beyond
Artetxe and Schwenk (2019) | [URL](#)

Improving Parallel Sentence Mining for Low-Resource and Endangered Languages
Okabe et al. (2025) | [URL](#)

Parallel Data, Tools and Interfaces in OPUS
Tiedemann (2012) | [URL](#)

The OPUS-MT Dashboard: A Toolkit for a Systematic Evaluation of Open Machine Translation Models
Tiedemann and de Gibert (2023) | [URL](#)

Improving Neural Machine Translation Models with Monolingual Data
Sennrich et al. (2016a) | [URL](#)

Understanding Back-Translation at Scale
Edunov et al. (2018) | [URL](#)

Exploring Diversity in Back Translation for Low-Resource NMT
Burchell et al. (2022) | [URL](#)

Beyond Dataset Creation: Critical View of Annotation Variation and Bias Probing of a Dataset for Online Radical Content Detection
Riabi et al. (2025) | [URL](#)

Treebanking user-generated content: a UD based overview of guidelines, corpora and unified recommendations
Sanguinetti et al. (2023) | [URL](#)

Enriching the NArabizi Treebank: A Multifaceted Approach to Supporting an Under-Resourced Language
Riabi et al. (2023) | [URL](#)

SUB-WORD MODELING & REPRESENTATION

“How to represent language when standard tokenization fails?”

SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing
Kudo and Richardson (2018) | [URL](#)

Unsupervised Multilingual Learning for POS Tagging
Snyder et al. (2008) | [URL](#)

Unsupervised Learning of the Morphology of a Natural Language
Goldsmith (2001) | [URL](#)

Neural Machine Translation of Rare Words with Subword Units
Sennrich et al. (2016b) | [URL](#)

Modeling Language Variation and Universals: A Survey on Typological Linguistics for Natural Language Processing
Ponti et al. (2019) | [URL](#)

Saliency-driven Word Alignment Interpretation for Neural Machine Translation
Ding et al. (2019) | [URL](#)

Selective Sharing for Multilingual Dependency Parsing
Naseem et al. (2012) | [URL](#)

Neural Machine Translation of Rare Words with Subword Units
Sennrich et al. (2016b) | [URL](#)

LINSPECTOR: Multilingual Probing Tasks for Word Representations
Şahin et al. (2020) | [URL](#)

Zero-Shot Neural Machine Translation with Self-Learning Cycle
Lakew et al. (2021) | [URL](#)

A Rule-Based Kurdish Text Transliteration System
Ahmadi (2019) | [URL](#)

Measuring Similarity by Linguistic Features rather than Frequency
Delmonte and Busetto (2022) | [URL](#)

EVALUATION & QUALITY ESTIMATION

“How to measure MT quality reliably when references and benchmarks are scarce?”

BLEU: A Method for Automatic Evaluation of Machine Translation
Papineni et al. (2002) | [URL](#)

A Structured Review of the Validity of BLEU
Reiter (2018) | [URL](#)

BLEURT: Learning Robust Metrics for Text Generation
Sellam et al. (2020) | [URL](#)

COMET: A Neural Framework for MT Evaluation
Rei et al. (2020) | [URL](#)

DOMAIN ADAPTATION & MOADLITIES

“What happens to machine translation when the language was never primarily a text?”

Analyzing phonetic variation in the traditional English dialects: Simultaneously clustering dialects and phonetic features

Wieling et al. (2013) | [URL](#)

LEARNING PARADIGMS & ARCHITECTURAL METHODS

“How to train or adapt models effectively with minimal supervision?”

Google’s Multilingual Neural Machine Translation System: Enabling Zero-Shot Translation

Johnson et al. (2017) | [URL](#)

Massively Multilingual Neural Machine Translation

Aharoni et al. (2019) | [URL](#)

Unsupervised Cross-lingual Representation Learning at Scale

Conneau et al. (2020) | [URL](#)

No Language Left Behind: Scaling Human-Centered Machine Translation

Team et al. (2022) | [URL](#)

Transfer Learning across Low-Resource, Related Languages for Neural Machine Translation

Nguyen and Chiang (2017) | [URL](#)

Polylingual Topic Models

Mimno et al. (2009) | [URL](#)

Ensemble Learning for Multi-source Neural Machine Translation

Garmash and Monz (2016) | [URL](#)

Attention is All You Need

Vaswani et al. (2017) | [URL](#)

Neural Machine Translation

Koehn (2020) | [URL](#)

Unsupervised Neural Machine Translation

Artetxe et al. (2018) | [URL](#)

Neural Machine Translation through Active Learning on Low-Resource Languages: The Case of Spanish to Mapudungun

Pendas et al. (2023) | [URL](#)

LLMs in the Loop: Leveraging Large Language Model Annotations for Active Learning in Low-Resource Languages

Kholodna et al. (2024) | [URL](#)

Zero-Shot Neural Machine Translation with Self-Learning Cycle

Lakew et al. (2021) | [URL](#)

Ensemble Learning for Multi-source Neural Machine Translation

Garmash and Monz (2016) | [URL](#)

A Survey of Domain Adaptation for Neural Machine Translation

Chu and Wang (2018) | [URL](#)

Finding the Right Recipe for Low Resource Domain Adaptation in Neural Machine Translation

Adams et al. (2022) | [URL](#)

EMERGING DIRECTIONS & OPEN PROBLEMS

“What new paradigms and unresolved issues will shape the near future?”

Should All Cross-Lingual Embeddings Speak English?

Anastasopoulos and Neubig (2020) | [URL](#)

Natural Language Processing for Dialects of a Language: A Survey

Joshi et al. (2025) | [URL](#)

Large Multimodal Models for Low-Resource Languages: A Survey

Lupascu et al. (2025) | [URL](#)

Neural Sign Language Translation

Camgoz et al. (2018) | [URL](#)

The Tatoeba Translation Challenge - Realistic Data Sets for Low Resource and Multilingual MT

Tiedemann (2020) | [URL](#)

Low-Resource Speech Translation

Bansal (2019) | [URL](#)

Sign Language Translation: A Survey of Approaches and Techniques

Liang et al. (2023) | [URL](#)

Speech Translation and the End-to-End Promise: Taking Stock of Where We Are

Sperber and Paulik (2020) | [URL](#)

A Practical Tool to Help Automate Interlinear Glossing: a Study on Mukr Kurdish

Asadpour et al. (2025) | [URL](#)

Learning to Recognize Dialect Features

Demszky et al. (2021) | [URL](#)

Multi-VALUE: A Framework for Cross-Dialectal English NLP

Ziems et al. (2023) | [URL](#)

Translation Asymmetry in LLMs as a Data Augmentation Factor: A Case Study for 6 Romansh Language Varieties

Vamvas et al. (2026) | [URL](#)

Versteasch du mi? Computational and Socio-Linguistic Perspectives on GenAI, LLMs, and Non-Standard Language

Platzgummer et al. (2026) | [URL](#)

Can Character-based Language Models Improve Downstream Task Performances In Low-Resource And Noisy Language Scenarios?

Riabi et al. (2021) | [URL](#)

Can Multilingual Language Models Transfer to an Unseen Dialect? A Case Study on North African Arabizi

Muller et al. (2020) | [URL](#)

Large Multimodal Models for Low-Resource Languages: A Survey

Lupascu et al. (2025) | [URL](#)

Additional Reading

Proceedings of the First Workshop on NLP applications to field linguistics
[Serikov et al. \(2022\)](#) | [URL](#)

Proceedings of the Second Workshop on NLP Applications to Field Linguistics
[Serikov et al. \(2023\)](#) | [URL](#)

Proceedings of the Third Workshop on NLP Applications to Field Linguistics
[Serikov et al. \(2024\)](#) | [URL](#)

Proceedings of the Fourth Workshop on NLP Applications to Field Linguistics
[Le Ferrand et al. \(2025\)](#) | [URL](#)

REFERENCES

- Virginia Adams, Sandeep Subramanian, Mike Chrzanowski, Oleksii Hrinchuk, and Oleksii Kuchaiev. 2022. Finding the Right Recipe for Low Resource Domain Adaptation in Neural Machine Translation. *arXiv:2206.01137* [cs] [doi:10.48550/arXiv.2206.01137](https://doi.org/10.48550/arXiv.2206.01137) 6
- Roei Aharoni, Melvin Johnson, and Orhan Firat. 2019. Massively Multilingual Neural Machine Translation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Thamar Solorio (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 3874–3884. [doi:10.18653/v1/N19-1388](https://doi.org/10.18653/v1/N19-1388) 6
- Sina Ahmadi. 2019. A Rule-Based Kurdish Text Transliteration System. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 18, 2 (Jan. 2019), 18:1–18:8. [doi:10.1145/3278623](https://doi.org/10.1145/3278623) 5
- Antonios Anastasopoulos and Graham Neubig. 2020. Should All Cross-Lingual Embeddings Speak English?. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 8658–8679. [doi:10.18653/v1/2020.acl-main.766](https://doi.org/10.18653/v1/2020.acl-main.766) 7
- Mikel Artetxe, Gorka Labaka, Eneko Agirre, and Kyunghyun Cho. 2018. Unsupervised Neural Machine Translation. *arXiv:1710.11041* [cs] [doi:10.48550/arXiv.1710.11041](https://doi.org/10.48550/arXiv.1710.11041) 6
- Mikel Artetxe and Holger Schwenk. 2019. Massively Multilingual Sentence Embeddings for Zero-Shot Cross-Lingual Transfer and Beyond. *Transactions of the Association for Computational Linguistics* 7 (2019), 597–610. [doi:10.1162/tac1_a_00288](https://doi.org/10.1162/tac1_a_00288) 4
- Hiwa Asadpour, Shu Okabe, and Alexander Fraser. 2025. A Practical Tool to Help Automate Interlinear Glossing: A Study on Mukri Kurdish. In *Proceedings of the Fourth Workshop on NLP Applications to Field Linguistics*, Éric Le Ferrand, Elena Klyachko, Anna Postnikova, Tatiana Shavrina, Oleg Serikov, Ekaterina Voloshina, and Ekaterina Vylomova (Eds.). Association for Computational Linguistics, Vienna, Austria, 65–75. <https://aclanthology.org/2025.fieldmatters-1.6/> 7
- Kalika Bali, Monojit Choudhury, Sunayana Sitaram, and Vivek Seshadri. 2019. ELLORA: Enabling Low Resource Languages with Technology. In *UNESCO International Conference on Language Technologies for All (LT4All)*. <https://www.microsoft.com/en-us/research/publication/ellora-enabling-low-resource-languages-with-technology/> 3
- Sameer Bansal. 2019. Low-Resource Speech Translation. (Dec. 2019). [doi:10.7488/era/86](https://doi.org/10.7488/era/86) 7
- Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? . In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. Association for Computing Machinery, New York, NY, USA, 610–623. [doi:10.1145/3442188.3445922](https://doi.org/10.1145/3442188.3445922) 3, 7
- Steven Bird. 2020. Decolonising Speech and Language Technology. In *Proceedings of the 28th International Conference on Computational Linguistics*. International Committee on Computational Linguistics, Barcelona, Spain (Online), 3504–3519. [doi:10.18653/v1/2020.coling-main.313](https://doi.org/10.18653/v1/2020.coling-main.313) 3
- Steven Bird. 2022. Local Languages, Third Spaces, and Other High-Resource Scenarios. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 7817–7829. [doi:10.18653/v1/2022.acl-long.539](https://doi.org/10.18653/v1/2022.acl-long.539) 3
- Steven Bird and Gary Simons. 2002. Seven Dimensions of Portability for Language Documentation and Description. *CoRR* cs.CL/0204020 (April 2002). [doi:10.1353/lan.2003.0149](https://doi.org/10.1353/lan.2003.0149) 4

- Laurie Burchell, Alexandra Birch, and Kenneth Heafield. 2022. Exploring Diversity in Back Translation for Low-Resource Machine Translation. In *Proceedings of the Third Workshop on Deep Learning for Low-Resource Natural Language Processing*. 67–79. arXiv:2206.00564 [cs] [doi:10.18653/v1/2022.deeplo-1.8](https://doi.org/10.18653/v1/2022.deeplo-1.8) 4
- Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. 2018. Neural Sign Language Translation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 7784–7793. https://openaccess.thecvf.com/content_cvpr_2018/html/Camgoz_Neural_Sign_Language_CVPR_2018_paper.html 7
- Chenhui Chu and Rui Wang. 2018. A Survey of Domain Adaptation for Neural Machine Translation. In *Proceedings of the 27th International Conference on Computational Linguistics*, Emily M. Bender, Leon Derczynski, and Pierre Isabelle (Eds.). Association for Computational Linguistics, Santa Fe, New Mexico, USA, 1304–1319. <https://aclanthology.org/C18-1111/6>
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised Cross-Lingual Representation Learning at Scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 8440–8451. [doi:10.18653/v1/2020.acl-main.747](https://doi.org/10.18653/v1/2020.acl-main.747) 6
- David Crystal. 2000. *Language Death*. Cambridge University Press, Cambridge. [doi:10.1017/CB09781139106856](https://doi.org/10.1017/CB09781139106856) 3
- Rodolfo Delmonte and Nicolò Busetto. 2022. Measuring Similarity by Linguistic Features Rather than Frequency. In *Proceedings of the 18th Joint ACL - ISO Workshop on Interoperable Semantic Annotation within LREC2022*, Harry Bunt (Ed.). European Language Resources Association, Marseille, France, 42–52. <https://aclanthology.org/2022.isa-1.6/> 5
- Dorottya Demszky, Devyani Sharma, Jonathan Clark, Vinodkumar Prabhakaran, and Jacob Eisenstein. 2021. Learning to Recognize Dialect Features. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Online, 2315–2338. [doi:10.18653/v1/2021.naacl-main.184](https://doi.org/10.18653/v1/2021.naacl-main.184) 7
- Shuoyang Ding, Hainan Xu, and Philipp Koehn. 2019. Saliency-Driven Word Alignment Interpretation for Neural Machine Translation. In *Proceedings of the Fourth Conference on Machine Translation (Volume 1: Research Papers)*, Ondřej Bojar, Rajen Chatterjee, Christian Federmann, Mark Fishel, Yvette Graham, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, André Martins, Christof Monz, Matteo Negri, Aurélie Névél, Mariana Neves, Matt Post, Marco Turchi, and Karin Verspoor (Eds.). Association for Computational Linguistics, Florence, Italy, 1–12. [doi:10.18653/v1/W19-5201](https://doi.org/10.18653/v1/W19-5201) 5
- Sergey Edunov, Myle Ott, Michael Auli, and David Grangier. 2018. Understanding Back-Translation at Scale. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 489–500. [doi:10.18653/v1/D18-1045](https://doi.org/10.18653/v1/D18-1045) 4
- Ekaterina Garmash and Christof Monz. 2016. Ensemble Learning for Multi-Source Neural Machine Translation. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, Yuji Matsumoto and Rashmi Prasad (Eds.). The COLING 2016 Organizing Committee, Osaka, Japan, 1409–1418. <https://aclanthology.org/C16-1133/> 6
- John Goldsmith. 2001. Unsupervised Learning of the Morphology of a Natural Language. *Computational Linguistics* 27, 2 (2001), 153–198. [doi:10.1162/089120101750300490](https://doi.org/10.1162/089120101750300490) 5
- Barry Haddow, Rachel Bawden, Antonio Valerio Miceli Barone, Jindřich Helcl, and Alexandra Birch. 2022. Survey of Low-Resource Machine Translation. *Computational Linguistics* 48, 3 (Sept. 2022), 673–732. [doi:10.1162/coli_a_00446](https://doi.org/10.1162/coli_a_00446) 3

- Melvin Johnson, Mike Schuster, Quoc V. Le, Maxim Krikun, Yonghui Wu, Zhifeng Chen, Nikhil Thorat, Fernanda Viégas, Martin Wattenberg, Greg Corrado, Macduff Hughes, and Jeffrey Dean. 2017. Google’s Multilingual Neural Machine Translation System: Enabling Zero-Shot Translation. *Transactions of the Association for Computational Linguistics* 5 (2017), 339–351. doi:10.1162/tac1_a_00065 6
- Aditya Joshi, Raj Dabre, Diptesh Kanojia, Zhuang Li, Haolan Zhan, Gholamreza Haffari, and Doris Dippold. 2025. Natural Language Processing for Dialects of a Language: A Survey. *ACM Comput. Surv.* 57, 6 (Feb. 2025), 149:1–149:37. doi:10.1145/3712060 7
- Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The State and Fate of Linguistic Diversity and Inclusion in the NLP World. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 6282–6293. doi:10.18653/v1/2020.acl-main.560 3
- Nataliia Kholodna, Sahib Julka, Mohammad Khodadadi, Muhammed Nurullah Gumus, and Michael Granitzer. 2024. LLMs in the Loop: Leveraging Large Language Model Annotations for Active Learning in Low-Resource Languages. arXiv:2404.02261 [cs] doi:10.48550/arXiv.2404.02261 6
- Philipp Koehn. 2005. Europarl: A Parallel Corpus for Statistical Machine Translation. In *Proceedings of Machine Translation Summit X: Papers*. Phuket, Thailand, 79–86. https://aclanthology.org/2005.mtsummit-papers.11/ 4
- Philipp Koehn. 2020. *Neural Machine Translation*. Cambridge University Press, Cambridge. doi:10.1017/9781108608480 6
- Philipp Koehn, Huda Khayrallah, Kenneth Heafield, and Mikel L. Forcada. 2018. Findings of the WMT 2018 Shared Task on Parallel Corpus Filtering. In *Proceedings of the Third Conference on Machine Translation: Shared Task Papers*, Ondřej Bojar, Rajen Chatterjee, Christian Federmann, Mark Fishel, Yvette Graham, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, Christof Monz, Matteo Negri, Aurélie Névél, Mariana Neves, Matt Post, Lucia Specia, Marco Turchi, and Karin Verspoor (Eds.). Association for Computational Linguistics, Belgium, Brussels, 726–739. doi:10.18653/v1/W18-6453 4
- András Kornai. 2013. Digital Language Death. *PLOS ONE* 8, 10 (Oct. 2013), e77056. doi:10.1371/journal.pone.0077056 2, 3
- Taku Kudo and John Richardson. 2018. SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Eduardo Blanco and Wei Lu (Eds.). Association for Computational Linguistics, Brussels, Belgium, 66–71. doi:10.18653/v1/D18-2012 4
- Surafel M. Lakew, Matteo Negri, and Marco Turchi. 2021. Zero-Shot Neural Machine Translation with Self-Learning Cycle. In *Proceedings of the 4th Workshop on Technologies for MT of Low Resource Languages (LoResMT2021)*, John Ortega, Atul Kr. Ojha, Katharina Kann, and Chao-Hong Liu (Eds.). Association for Machine Translation in the Americas, Virtual, 96–113. https://aclanthology.org/2021.mtsummit-loresmt.10/ 5, 6
- Théo Lasnier, Armel Zebaze, Djamé Seddah, Rachel Bawden, and Benoît Sagot. 2026. Disentangling Meaning from Language in LLM-Based Machine Translation. arXiv:2602.04613 [cs] doi:10.48550/arXiv.2602.04613 8
- Éric Le Ferrand, Elena Klyachko, Anna Postnikova, Tatiana Shavrina, Oleg Serikov, Ekaterina Voloshina, and Ekaterina Vylomova (Eds.). 2025. *Proceedings of the Fourth Workshop on NLP Applications to Field Linguistics*. Association for Computational Linguistics, Vienna, Austria. https://aclanthology.org/2025.fieldmatters-1.0/ 8
- Zeyu Liang, Huailing Li, and Jianping Chai. 2023. Sign Language Translation: A Survey of Approaches and Techniques. *Electronics* 12, 12 (Jan. 2023), 2678. doi:10.3390/electronics12122678 7

- Zoey Liu, Crystal Richardson, Richard Hatcher, and Emily Prud'hommeaux. 2022. Not Always about You: Prioritizing Community Needs When Developing Endangered Language Technology. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 3933–3944. doi:10.18653/v1/2022.acl-long.272 3
- Marian Lupascu, Ana-Cristina Rogoz, Mihai Sorin Stupariu, and Radu Tudor Ionescu. 2025. Large Multimodal Models for Low-Resource Languages: A Survey. arXiv:2502.05568 [cs] doi:10.48550/arXiv.2502.05568 7
- Manuel Mager, Ximena Gutierrez-Vasques, Gerardo Sierra, and Ivan Meza-Ruiz. 2018. Challenges of Language Technologies for the Indigenous Languages of the Americas. In *Proceedings of the 27th International Conference on Computational Linguistics*, Emily M. Bender, Leon Derczynski, and Pierre Isabelle (Eds.). Association for Computational Linguistics, Santa Fe, New Mexico, USA, 55–69. <https://aclanthology.org/C18-1006> 3
- David Mimno, Hanna M. Wallach, Jason Naradowsky, David A. Smith, and Andrew McCallum. 2009. Polylingual Topic Models. In *Proceedings of the 2009 Conference on Empirical Methods in Natural Language Processing*, Philipp Koehn and Rada Mihalcea (Eds.). Association for Computational Linguistics, Singapore, 880–889. <https://aclanthology.org/D09-1092/> 6
- Christopher Moseley and Alexandre Nicolas. 2010. *Atlas of the World's Languages in Danger* (3rd ed., entirely rev., enl., upd. ed.). Paris : UNESCO, 2010. <https://unesdoc.unesco.org/ark:/48223/pf0000187026> 3
- Benjamin Muller, Benoit Sagot, and Djamé Seddah. 2020. Can Multilingual Language Models Transfer to an Unseen Dialect? A Case Study on North African Arabizi. arXiv:2005.00318 [cs] doi:10.48550/arXiv.2005.00318 7
- Tahira Naseem, Regina Barzilay, and Amir Globerson. 2012. Selective Sharing for Multilingual Dependency Parsing. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Haizhou Li, Chin-Yew Lin, Miles Osborne, Gary Geunbae Lee, and Jong C. Park (Eds.). Association for Computational Linguistics, Jeju Island, Korea, 629–637. <https://aclanthology.org/P12-1066/> 5
- Toan Q. Nguyen and David Chiang. 2017. Transfer Learning across Low-Resource, Related Languages for Neural Machine Translation. In *Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, Greg Kondrak and Taro Watanabe (Eds.). Asian Federation of Natural Language Processing, Taipei, Taiwan, 296–301. <https://aclanthology.org/I17-2050/> 6
- Shu Okabe, Katharina Hämmerl, and Alexander Fraser. 2025. Improving Parallel Sentence Mining for Low-Resource and Endangered Languages. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (Eds.). Association for Computational Linguistics, Vienna, Austria, 196–205. doi:10.18653/v1/2025.acl-short.17 4
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: A Method for Automatic Evaluation of Machine Translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, Pierre Isabelle, Eugene Charniak, and Dekang Lin (Eds.). Association for Computational Linguistics, Philadelphia, Pennsylvania, USA, 311–318. doi:10.3115/1073083.1073135 5
- Begoña Pendas, Andres Carvallo, and Carlos Aspillaga. 2023. Neural Machine Translation through Active Learning on Low-Resource Languages: The Case of Spanish to Mapudungun. In *Proceedings of the Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP)*, Manuel Mager, Abteen Ebrahimi, Arturo Oncevay, Enora Rice, Shruti Rijhwani, Alexis Palmer, and Katharina Kann (Eds.). Association for Computational Linguistics, Toronto, Canada, 6–11. doi:10.18653/v1/2023.americasnlp-1.2 6

- Daniel Pimienta and Gilvan Müller de Oliveira. 2022. Cyber-Geography of Languages: Part 1: Method, Results and Focus on English. *The International Review of Information Ethics* 32, 1 (Nov. 2022). doi:10.29173/irrie488 3
- Verena Platzgummer, John McCrae, and Sina Ahmadi. 2026. \textit{Versteasch Du Mi?} Computational and Socio-Linguistic Perspectives on GenAI, LLMs, and Non-Standard Language. <https://arxiv.org/abs/2603.28213v1> 7
- Edoardo Maria Ponti, Helen O’Horan, Yevgeni Berzak, Ivan Vulić, Roi Reichart, Thierry Poibeau, Ekaterina Shutova, and Anna Korhonen. 2019. Modeling Language Variation and Universals: A Survey on Typological Linguistics for Natural Language Processing. *Computational Linguistics* 45, 3 (Sept. 2019), 559–601. doi:10.1162/coli_a_00357 5
- Ricardo Rei, Craig Stewart, Ana C Farinha, and Alon Lavie. 2020. COMET: A Neural Framework for MT Evaluation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, Online, 2685–2702. doi:10.18653/v1/2020.emnlp-main.213 5
- Ehud Reiter. 2018. A Structured Review of the Validity of BLEU. *Computational Linguistics* 44, 3 (Sept. 2018), 393–401. doi:10.1162/coli_a_00322 5
- Arij Riabi, Menel Mahamdi, and Djamé Seddah. 2023. Enriching the NArabizi Treebank: A Multifaceted Approach to Supporting an Under-Resourced Language. In *Proceedings of the 17th Linguistic Annotation Workshop (LAW-XVII)*, Jakob Prange and Annemarie Friedrich (Eds.). Association for Computational Linguistics, Toronto, Canada, 266–278. doi:10.18653/v1/2023.law-1.26 2, 4
- Arij Riabi, Virginie Moulleron, Menel Mahamdi, Wissam Antoun, and Djamé Seddah. 2025. Beyond Dataset Creation: Critical View of Annotation Variation and Bias Probing of a Dataset for Online Radical Content Detection. In *Proceedings of the 31st International Conference on Computational Linguistics*, Owen Rambow, Leo Wanner, Marianna Apidianaki, Hend Al-Khalifa, Barbara Di Eugenio, and Steven Schockaert (Eds.). Association for Computational Linguistics, Abu Dhabi, UAE, 8640–8663. <https://aclanthology.org/2025.coling-main.578/> 4
- Arij Riabi, Benoît Sagot, and Djamé Seddah. 2021. Can Character-Based Language Models Improve Downstream Task Performances In Low-Resource And Noisy Language Scenarios?. In *Proceedings of the Seventh Workshop on Noisy User-Generated Text (W-NUT 2021)*, Wei Xu, Alan Ritter, Tim Baldwin, and Afshin Rahimi (Eds.). Association for Computational Linguistics, Online, 423–436. doi:10.18653/v1/2021.wnut-1.47 7
- Gözde Gül Şahin, Clara Vania, Ilia Kuznetsov, and Iryna Gurevych. 2020. LINSPECTOR: Multilingual Probing Tasks for Word Representations. *Computational Linguistics* 46, 2 (June 2020), 335–385. doi:10.1162/coli_a_00376 5
- Manuela Sanguinetti, Cristina Bosco, Lauren Cassidy, Özlem Çetinoğlu, Alessandra Teresa Cignarella, Teresa Lynn, Ines Rehbein, Josef Ruppenhofer, Djamé Seddah, and Amir Zeldes. 2023. Treebanking User-Generated Content: A UD Based Overview of Guidelines, Corpora and Unified Recommendations. *Language Resources and Evaluation* 57, 2 (June 2023), 493–544. doi:10.1007/s10579-022-09581-9 4
- Holger Schwenk, Guillaume Wenzek, Sergey Edunov, Edouard Grave, Armand Joulin, and Angela Fan. 2021. CCMatrix: Mining Billions of High-Quality Parallel Sentences on the Web. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, Online, 6490–6500. doi:10.18653/v1/2021.acl-long.507 4
- Thibault Sellam, Dipanjan Das, and Ankur Parikh. 2020. BLEURT: Learning Robust Metrics for Text Generation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 7881–7892. doi:10.18653/v1/2020.acl-main.704 5

- Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016a. Improving Neural Machine Translation Models with Monolingual Data. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Katrin Erk and Noah A. Smith (Eds.). Association for Computational Linguistics, Berlin, Germany, 86–96. [doi:10.18653/v1/P16-1009](https://doi.org/10.18653/v1/P16-1009) 4
- Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016b. Neural Machine Translation of Rare Words with Subword Units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Katrin Erk and Noah A. Smith (Eds.). Association for Computational Linguistics, Berlin, Germany, 1715–1725. [doi:10.18653/v1/P16-1162](https://doi.org/10.18653/v1/P16-1162) 5
- Oleg Serikov, Ekaterina Voloshina, Anna Postnikova, Elena Klyachko, Ekaterina Neminova, Ekaterina Vylomova, Tatiana Shavrina, Eric Le Ferrand, Valentin Malykh, Francis Tyers, Timofey Arkhangelskiy, Vladislav Mikhailov, and Alena Fenogenova (Eds.). 2022. *Proceedings of the First Workshop on NLP Applications to Field Linguistics*. International Conference on Computational Linguistics, Gyeongju, Republic of Korea. <https://aclanthology.org/2022.fieldmatters-1.0/> 8
- Oleg Serikov, Ekaterina Voloshina, Anna Postnikova, Elena Klyachko, Ekaterina Vylomova, Tatiana Shavrina, Eric Le Ferrand, Valentin Malykh, Francis Tyers, Timofey Arkhangelskiy, and Vladislav Mikhailov (Eds.). 2023. *Proceedings of the Second Workshop on NLP Applications to Field Linguistics*. Association for Computational Linguistics, Dubrovnik, Croatia. <https://aclanthology.org/2023.fieldmatters-1.0/> 8
- Oleg Serikov, Ekaterina Voloshina, Anna Postnikova, Saliha Muradoglu, Eric Le Ferrand, Elena Klyachko, Ekaterina Vylomova, Tatiana Shavrina, and Francis Tyers (Eds.). 2024. *Proceedings of the Third Workshop on NLP Applications to Field Linguistics*. Association for Computational Linguistics, Bangkok, Thailand. <https://aclanthology.org/2024.fieldmatters-1.0/> 8
- Benjamin Snyder, Tahira Naseem, Jacob Eisenstein, and Regina Barzilay. 2008. Unsupervised Multilingual Learning for POS Tagging. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, Mirella Lapata and Hwee Tou Ng (Eds.). Association for Computational Linguistics, Honolulu, Hawaii, 1041–1050. <https://aclanthology.org/D08-1109/> 4
- Matthias Sperber and Matthias Paulik. 2020. Speech Translation and the End-to-End Promise: Taking Stock of Where We Are. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 7409–7421. [doi:10.18653/v1/2020.acl-main.661](https://doi.org/10.18653/v1/2020.acl-main.661) 3, 7
- NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loic Barrault, Gabriel Mejia Gonzalez, Prangthip Hansanti, John Hoffman, Semarley Jarrett, Kaushik Ram Sadagopan, Dirk Rowe, Shannon Spruit, Chau Tran, Pierre Andrews, Necip Fazil Ayan, Shruti Bhosale, Sergey Edunov, Angela Fan, Cynthia Gao, Vedanuj Goswami, Francisco Guzmán, Philipp Koehn, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, and Jeff Wang. 2022. No Language Left Behind: Scaling Human-Centered Machine Translation. arXiv:2207.04672 [cs] [doi:10.48550/arXiv.2207.04672](https://doi.org/10.48550/arXiv.2207.04672) 6
- Jörg Tiedemann. 2012. Parallel Data, Tools and Interfaces in OPUS. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC'12)*. European Language Resources Association (ELRA), Istanbul, Turkey, 2214–2218. http://www.lrec-conf.org/proceedings/lrec2012/pdf/463_Paper.pdf 4
- Jörg Tiedemann. 2020. The Tatoeba Translation Challenge – Realistic Data Sets for Low Resource and Multilingual MT. In *Proceedings of the Fifth Conference on Machine Translation*, Loïc Barrault, Ondřej Bojar, Fethi Bougares, Rajen Chatterjee, Marta R. Costa-

- jussà, Christian Federmann, Mark Fishel, Alexander Fraser, Yvette Graham, Paco Guzman, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, André Martins, Makoto Morishita, Christof Monz, Masaaki Nagata, Toshiaki Nakazawa, and Matteo Negri (Eds.). Association for Computational Linguistics, Online, 1174–1182. <https://aclanthology.org/2020.wmt-1.139/> 7
- Jörg Tiedemann and Ona de Gibert. 2023. The OPUS-MT Dashboard – A Toolkit for a Systematic Evaluation of Open Machine Translation Models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, Danushka Bollegala, Ruihong Huang, and Alan Ritter (Eds.). Association for Computational Linguistics, Toronto, Canada, 315–327. [doi:10.18653/v1/2023.acl-demo.30](https://doi.org/10.18653/v1/2023.acl-demo.30) 4
- Jannis Vamvas, Ignacio Pérez Prat, Angela Heldstab, Dominic P. Fischer, Sina Ahmadi, and Rico Sennrich. 2026. Translation Asymmetry in LLMs as a Data Augmentation Factor: A Case Study for 6 Romansh Language Varieties. <https://arxiv.org/abs/2603.25489v1> 7
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. *arXiv:1706.03762* [cs] [doi:10.48550/arXiv.1706.03762](https://doi.org/10.48550/arXiv.1706.03762) 6
- Martijn Wieling, Robert G. Shackleton, Jr., and John Nerbonne. 2013. Analyzing Phonetic Variation in the Traditional English Dialects: Simultaneously Clustering Dialects and Phonetic Features. *Literary and Linguistic Computing* 28, 1 (April 2013), 31–41. [doi:10.1093/l1c/fqs047](https://doi.org/10.1093/l1c/fqs047) 6
- Caleb Ziems, William Held, Jingfeng Yang, Jwala Dhamala, Rahul Gupta, and Diyi Yang. 2023. Multi-VALUE: A Framework for Cross-Dialectal English NLP. *arXiv:2212.08011* [cs] [doi:10.48550/arXiv.2212.08011](https://doi.org/10.48550/arXiv.2212.08011) 7